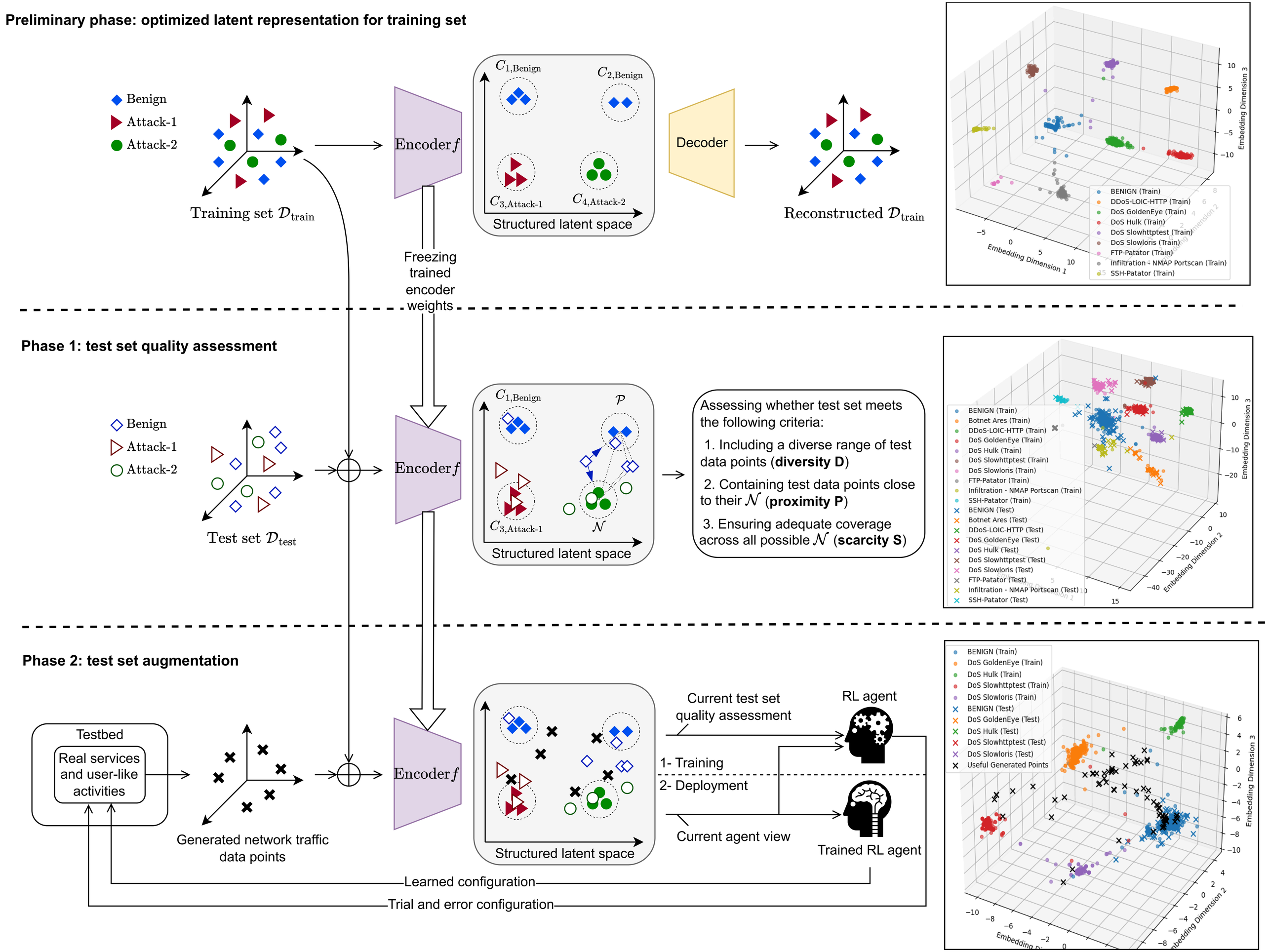
Preliminary phase: optimized latent representation for training set
Benign
Attack-1
Attack-2
Training set $\mathcal{D}_{\text{train}}$
Encoder f
$C_{1,\text{Benign}}$
$C_{2,\text{Benign}}$
$C_{3,\text{Attack-1}}$
$C_{4,\text{Attack-2}}$
Structured latent space
Decoder
Reconstructed $\mathcal{D}_{\text{train}}$
Freezing trained encoder weights
Embedding Dimension 1
Embedding Dimension 2
Embedding Dimension 3
BENIGN (Train)
DDoS-LOIC-HTTP (Train)
DoS GoldenEye (Train)
DoS Hulk (Train)
DoS Slowhttptest (Train)
DoS Slowloris (Train)
FTP-Patator (Train)
Infiltration - NMAP Portscan (Train)
SSH-Patator (Train)
Phase 1: test set quality assessment
Benign
Attack-1
Attack-2
Test set $\mathcal{D}_{\text{test}}$
Encoder f
$C_{1,\text{Benign}}$
$\mathcal{P}$
$C_{3,\text{Attack-1}}$
$\mathcal{N}$
Structured latent space
Assessing whether test set meets the following criteria:
1. Including a diverse range of test data points (**diversity D**)
2. Containing test data points close to their $\mathcal{N}$ (**proximity P**)
3. Ensuring adequate coverage across all possible $\mathcal{N}$ (**scarcity S**)
BENIGN (Train)
Botnet Ares (Train)
DDoS-LOIC-HTTP (Train)
DoS GoldenEye (Train)
DoS Hulk (Train)
DoS Slowhttptest (Train)
DoS Slowloris (Train)
FTP-Patator (Train)
Infiltration - NMAP Portscan (Train)
SSH-Patator (Train)
BENIGN (Test)
Botnet Ares (Test)
DDoS-LOIC-HTTP (Test)
DoS GoldenEye (Test)
DoS Hulk (Test)
DoS Slowhttptest (Test)
DoS Slowloris (Test)
FTP-Patator (Test)
Infiltration - NMAP Portscan (Test)
SSH-Patator (Test)
Phase 2: test set augmentation
Testbed
Real services and user-like activities
Generated network traffic data points
Encoder f
Structured latent space
Current test set quality assessment
RL agent
1- Training
2- Deployment
Current agent view
Trained RL agent
Learned configuration
Trial and error configuration
BENIGN (Train)
DoS GoldenEye (Train)
DoS Hulk (Train)
DoS Slowhttptest (Train)
DoS Slowloris (Train)
BENIGN (Test)
DoS GoldenEye (Test)
DoS Hulk (Test)
DoS Slowhttptest (Test)
DoS Slowloris (Test)
Useful Generated Points